🚀 Properties of Different Function Return Types in C++

Functions in C++ can return different types of values, each with its own properties and use cases. Let's explore int, bool, void, float, char, string, pointers, references, and more!

📌 1⃣ int (Integer Functions)

✅ Properties

✓ Used to return whole numbers (int).
✓ Can be used in arithmetic expressions.
✓ Can return negative and positive numbers.

✅ Example: Returning int

cpp

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```
#include <iostream>

using namespace std;


int add(int a, int b) {

   return a + b;  // Returns an integer

}


int main() {

   int result = add(5, 10);  // Stores returned value

   cout << "Sum: " << result << endl;

   return 0;

}
```

📝 Output

makefile

CopyEdit

Sum: 15

✔ Best Use Cases: Mathematical operations, counting loops.

📌 2⃣ bool (Boolean Functions)

✅ Properties

✔ Returns true (1) or false (0).
✔ Used for decision-making (if-else conditions).
✔ Often used in searching, checking conditions, and validation functions.

✅ Example: Checking if a Number is Even

cpp

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```
#include <iostream>

using namespace std;


bool isEven(int num) {

   return (num % 2 == 0);  // Returns true if even, false if odd

}


int main() {

   int number = 8;

   if (isEven(number)) {

      cout << number << " is even." << endl;

   } else {

      cout << number << " is odd." << endl;

   }
```

```
    return 0;
}
```

📝 Output

csharp

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8 is even.

✓ Best Use Cases: Condition checking, validation functions, flag variables.

📌 3️void (Functions That Don't Return a Value)

✅ Properties

✓ Does not return anything.
✓ Used for printing, modifying global variables, or performing an action.
✓ Cannot be used in expressions like int x = myFunction(); (because it doesn't return a value).

✅ Example: Printing a Message

cpp

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```
#include <iostream>
using namespace std;

void greet() {
    cout << "Hello, welcome to C++ programming!" << endl;
}

int main() {
    greet();  // Function call (doesn't return anything)
    return 0;
```

}

📝 Output

css

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Hello, welcome to C++ programming!

✓ Best Use Cases: Printing, modifying global variables, user interactions.

📌 4⃣ float & double (Floating-Point Functions)

✅ Properties

✓ Returns decimal numbers (floating-point).
✓ float has less precision, double has higher precision.
✓ Used in scientific calculations, measuring values.

✅ Example: Calculating the Area of a Circle

cpp

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```
#include <iostream>

using namespace std;


double areaOfCircle(double radius) {

    return 3.14159 * radius * radius;  // Returns double value

}


int main() {

    double area = areaOfCircle(5.5);

    cout << "Area: " << area << endl;

    return 0;
```

}

📝 Output

makefile

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Area: 95.0331

✓ Best Use Cases: Scientific calculations, measuring, statistics.

📌 5⃣ char (Character Functions)

✅ Properties

✓ Returns a single character ('A', 'B', '@').
✓ Useful in text-based processing.
✓ Can return ASCII characters.

✅ Example: Returning a Grade

cpp

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```
#include <iostream>

using namespace std;


char getGrade(int marks) {

    if (marks >= 90) return 'A';

    else if (marks >= 80) return 'B';

    else return 'C';

}


int main() {

    char grade = getGrade(85);
```

```
    cout << "Grade: " << grade << endl;
    return 0;
}
```

📝 Output

makefile

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Grade: B

✓ Best Use Cases: Character processing, grading systems.

📌 6⃣ string (Returning Strings)

✅ Properties

✓ Returns a sequence of characters (string).
✓ Used in handling text, user input, and file processing.

✅ Example: Returning a Full Name

cpp

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```
#include <iostream>
using namespace std;

string getFullName(string firstName, string lastName) {
    return firstName + " " + lastName;  // Concatenation
}

int main() {
    string name = getFullName("John", "Doe");
    cout << "Full Name: " << name << endl;
```

```
    return 0;
}
```

📝 Output

pgsql

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```
Full Name: John Doe
```

✓ Best Use Cases: Handling user input, file processing.

📌 7⃣ int* (Returning Pointers)

✅ Properties

✓ Returns memory addresses.
✓ Used in dynamic memory allocation.
✓ Can lead to dangling pointers if not handled properly.

✅ Example: Returning an Array

cpp

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```
#include <iostream>
using namespace std;

int* getArray() {
    static int arr[3] = {10, 20, 30};  // Static to avoid deletion
    return arr;
}

int main() {
    int* arr = getArray();
```

```
    cout << "First element: " << arr[0] << endl;
    return 0;
}
```

📝 Output

sql

CopyEdit

```
First element: 10
```

✓ Best Use Cases: Dynamic memory, passing arrays.

📌 8⃣ Returning struct or class Objects

✅ Properties

✓ Can return complex data structures.
✓ Useful in Object-Oriented Programming (OOP).

✅ Example: Returning a struct

cpp

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```
#include <iostream>
using namespace std;

struct Student {
    string name;
    int age;
};

Student getStudent() {
    return {"Alice", 20};  // Returning a struct
```

```
}

int main() {
    Student s = getStudent();
    cout << "Name: " << s.name << ", Age: " << s.age << endl;
    return 0;
}
```

📝 Output

yaml

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Name: Alice, Age: 20

✓ Best Use Cases: OOP, handling multiple values.

📌 9⃣ Returning by Reference (&)

✅ Properties

✓ Returns a reference to an existing variable.
✓ Avoids extra copying, making it faster.

✅ Example: Returning a Reference

cpp

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```
#include <iostream>
using namespace std;

int& getRef(int &x) {
    return x;
}
```

```
int main() {
    int num = 10;
    int &ref = getRef(num);
    ref = 20;
    cout << "Updated num: " << num << endl;
    return 0;
}
```

📝 Output

yaml

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Updated num: 20

✓ Best Use Cases: Optimizing performance, modifying variables.

🔥 Summary Table

Return Type	Properties	Best Use Cases
int	Returns whole numbers	Math, counting loops
bool	Returns true/false	Condition checking
void	No return value	Printing, modifying globals
float/double	Returns decimal numbers	Scientific calculations
char	Returns a single character	Character processing
string	Returns text	User input, file handling
int*	Returns a pointer	Dynamic memory
struct/class	Returns an object	OOP, complex data

Return Type	Properties	Best Use Cases
& (reference)	Returns original variable	Optimizing performance

Would you like deeper explanations on any type? 🚀 🔥